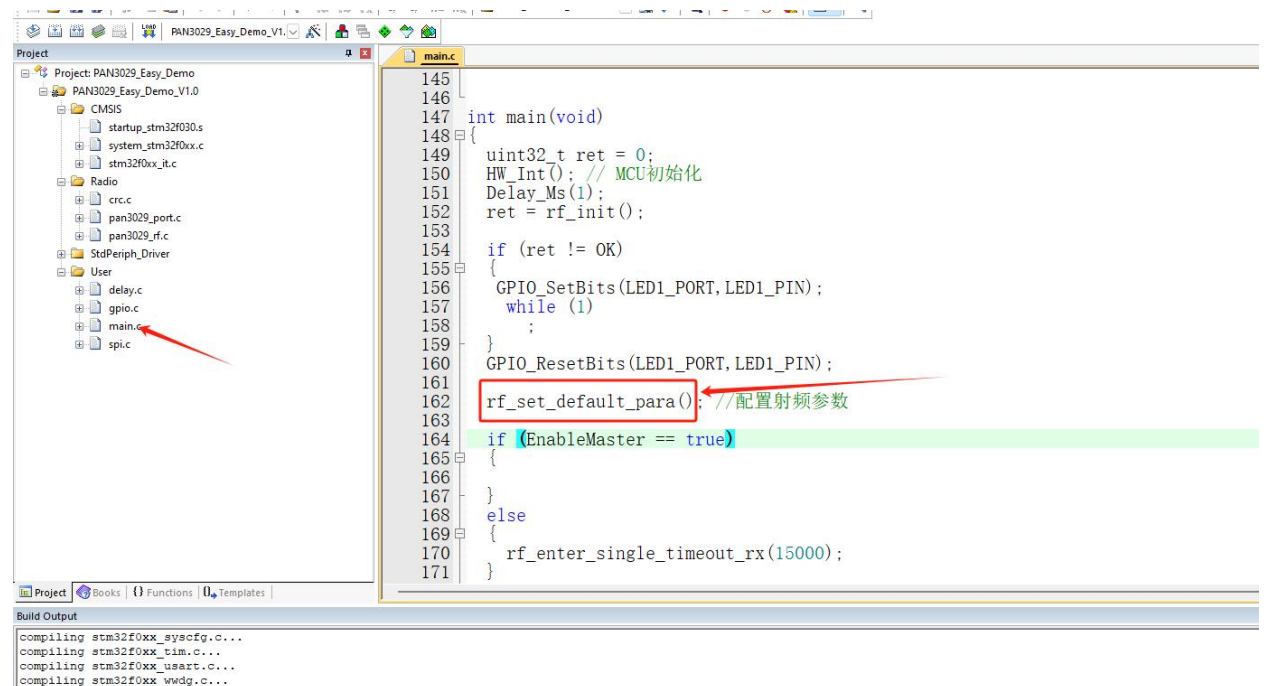


I. Modifying Parameters on the PAN3029 Side

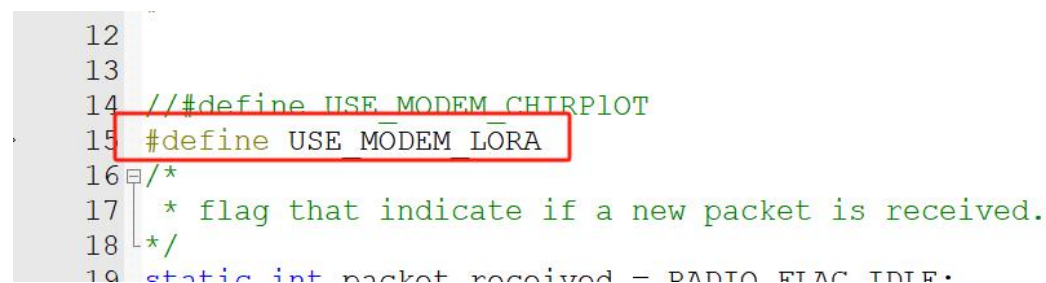
1. Open the PAN3029 project, as shown below:



2. Locate ``rf_set_default_para();`` // Configure RF parameters



3. Based on step 2, enable the macro definition ``USE_MODEM_LORA``, as shown below:



```

52 | // return result
53 | */
54 | RF_Err_t rf_set_default_para(void)
55 | {
56 |
57 | #if defined( USE_MODEM_LORA )
58 |     set_test_model_reg(); //LoRa模式
59 |     RF_ASSERT(rf_set_crc(CRC_OFF)); //关闭CRC
60 | #else
61 |     RF_ASSERT(rf_set_crc(CRC_ON));
62 | #endif
63 |
64 |     RF_ASSERT(rf_set_freq(DEFAULT_FREQ));
65 |     RF_ASSERT(rf_set_code_rate(DEFAULT_CR));
66 |     RF_ASSERT(rf_set_bw(DEFAULT_BW));
67 |     RF_ASSERT(rf_set_sf(DEFAULT_SF));
68 |     RF_ASSERT(rf_set_tx_power(DEFAULT_PWR));
69 |     return OK;
70 | }

```

II. Modifying Parameters on the SX1268 Side

1. Open the SX1268 project, as shown below:

 SX1268ZTR4-GC_Easy_Demo_v2.3 2024/7/3 10:38 folder

2. Open `main.c` and find the location shown below. You need to turn off CRC, i.e., change `true` to `false` in the `CrcOn` location, as shown below:

```

5 | // ***** USER CODE *****
6 | #if defined( USE_MODEM_LORA )
7 |
8 |     Radio.SetTxConfig( MODEM_LORA, TX_OUTPUT_POWER, 0, LORA_BANDWIDTH,
9 |                         LORA_SPREADING_FACTOR, LORA_CODINGRATE,
10 |                        LORA_PREAMBLE_LENGTH, LORA_FIX_LENGTH_PAYLOAD_ON,
11 |                        修改前 true, 0, 0, LORA_IQ_INVERSION_ON, 3000 );
12 |
13 |     Radio.SetRxConfig( MODEM_LORA, LORA_BANDWIDTH, LORA_SPREADING_FACTOR,
14 |                        LORA_CODINGRATE, 0, LORA_PREAMBLE_LENGTH,
15 |                        LORA_SYMBOL_TIMEOUT, LORA_FIX_LENGTH_PAYLOAD_ON,
16 |                        0, true, 0, 0, LORA_IQ_INVERSION_ON, false );
17 |

```

```

// #define USE_MODEM_FSK /
Radio.SetTxConfig( MODEM_LORA, TX_OUTPUT_POWER, 0, LORA_BANDWIDTH,
LORA_SPREADING_FACTOR, LORA_CODINGRATE,
修改后 false, 0, 0, LORA_IQ_INVERSION_ON, 3000 );

Radio.SetRxConfig( MODEM_LORA, LORA_BANDWIDTH, LORA_SPREADING_FACTOR,
LORA_CODINGRATE, 0, LORA_PREAMBLE_LENGTH,
LORA_SYMBOL_TIMEOUT, LORA_FIX_LENGTH_PAYLOAD_ON,
0, false, 0, 0, LORA_IQ_INVERSION_ON, false );
// #define USE_MODEM_FSK /

```

III. After modifying the above parameters, for interoperability, both sides also need to set the frequency, SF (spreading factor), BW (bandwidth), and CR to be consistent for communication, as shown below:

SX1268 Side:

Modify in `main.c`

```

27 // #define USE_MODEM_FSK
28
29 #define REGION_CN779
30
31 #if defined( REGION_AS923 )
32
33 #define RF_FREQUENCY 923000000 // Hz
34
35 #elif defined( REGION_AU915 )
36
37 #define RF_FREQUENCY 915000000 // Hz
38
39 #elif defined( REGION_CN779 )
40 FERQ=433.000000M
41 #define RF_FREQUENCY 433000000 // Hz
42
43 #elif defined( REGION_EU868 )
44
45 #define RF_FREQUENCY 868000000 // Hz
46
47 #elif defined( REGION_KR920 )
48
49 #define RF_FREQUENCY 920000000 // Hz
50
51 #elif defined( REGION_IN865 )
52
53 #define RF_FREQUENCY 865000000 // Hz

```

```

78 /*
79  * Radio events function pointer
80  */
81
82 static RadioEvents_t RadioEvents;
83
84 #if defined( USE_MODEM_LORA )
85
86 #define LORA_BANDWIDTH          BW=125K 0      // [0: 125 kHz,
87                                           // 1: 250 kHz,
88                                           // 2: 500 kHz,
89                                           // 3: Reserved]
90 #define LORA_SPREADING_FACTOR   SF=7 7        // [SF7..SF12]
91 #define LORA_CODINGRATE         CR=4/8 4       // [1: 4/5,
92                                           // 2: 4/6,
93                                           // 3: 4/7,
94                                           // 4: 4/8]
95 #define LORA_PREAMBLE_LENGTH    8            // Same for Tx and Rx
96 #define LORA_SYMBOL_TIMEOUT     0            // Symbols
97 #define LORA_FIX_LENGTH_PAYLOAD_ON false
98 #define LORA_IQ_INVERSION_ON    false
99
100 #elif defined( USE_MODEM_FSK )
101
102 #define FSK_FDEV                 5e3          // Hz
103 #define FSK_DATARATE             2.4e3       // bps
104 #define FSK_BANDWIDTH            20e3        // Hz >> DSB in sx126x
105 #define FSK_AFC_BANDWIDTH        100e3       // Hz
106 #define FSK_PREAMBLE_LENGTH      5           // Same for Tx and Rx
107 #define FSK_FIX_LENGTH_PAYLOAD_ON false

```

PAN3029 Side:

Modify in `radio.h`

```

pan3029_port.c  main.c  pan3029_rf.c  pan3029_rf.h  pan3029_port.h
15 #define ETSI_433
16 // #define ETSI_868
17
18 #if defined(JAP_915)
19 #define DEFAULT_PWR             17
20 #define DEFAULT_FREQ            (915000000)
21 #define DEFAULT_SF              SF_7
22 #define DEFAULT_BW              BW_125K
23 #define DEFAULT_CR              CODE_RATE_48
24
25 #elif defined(FCC_915)
26 #define DEFAULT_PWR             16
27 #define DEFAULT_FREQ            (915000000)
28 #define DEFAULT_SF              SF_7
29 #define DEFAULT_BW              BW_125K
30 #define DEFAULT_CR              CODE_RATE_48
31
32 #elif defined(ETSI_433)
33 #define DEFAULT_PWR             22
34 #define DEFAULT_FREQ            (433000000)
35 #define DEFAULT_SF              SF_7
36 #define DEFAULT_BW              BW_125K
37 #define DEFAULT_CR              CODE_RATE_48
38
39 #elif defined(ETSI_868)
40 #define DEFAULT_PWR             12
41 #define DEFAULT_FREQ            (868000000)
42 #define DEFAULT_SF              SF_7
43 #define DEFAULT_BW              BW_125K
44 #define DEFAULT_CR              CODE_RATE_48
45
46 #else
47 /*Default parameter configuration*/
48 #define DEFAULT_PWR             22
49 #define DEFAULT_FREQ            (470000000)
50 #define DEFAULT_SF              SF_5
51 #define DEFAULT_BW              BW_500K
52 #define DEFAULT_CR              CODE_RATE_45
53 #endif
54
55

```